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TGADHead: An efficient and accurate task-guided attention-decoupled head for single-stage object detection

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ABSTRACT

In object detection, localization and classification of the targets are two fundamental subtasks that underpin the application of many knowledge-based intelligent models in various visual fields. However, current methods show inconsistency between the two subtasks, i.e., accurate localization may show a poor classification score or vice versa, due to inadequate design of the detectors. This inconsistency significantly degrades the overall detection accuracy. To address this issue, we propose a novel Task-Guided Attention-Decoupled Head (TGADHead) to improve detection accuracy using an efficient single-stage approach. The proposed framework consists of two inter-connected components: Task Decoupled Attention Distributor (TDAD) and Task Correlation Network (TCN). In the first component, we propose TDAD with two well-designed task specific attention perceptrons to enhance the spatial information required for localization and the semantic information required for classification, respectively. This task specific prediction mechanism improves classification and location performance. Secondly, we construct a Task Correlation Network (TCN) for each of decoupled branch to transfer the correlation between classification and localization features while maintaining the consistency of the two subtasks, i.e., an accurate object prediction outcome should simultaneously have a high quality location boundary box and a high classification score. We achieve +1.9 Average Precision (AP) on the MS-COCO compared to the state-of-the-art detectors.

1. Introduction

Object detection aims to identify objects in regions of interest in natural or industrial scenes and is a major yet challenging task in computer vision [1–4]. In the last decade, significant progress in object detection has been made [5]. It has been successfully applied to video tracking [6–8], video understanding [9,10], industrial quality monitoring [11,12], pedestrian recognition [13,14] and many other fields. In the process of object recognition, it is not only necessary to sort it into the category to which the object belongs (classification), but also to identify the location of the region of interest (localization), and therefore it is usually regarded as a multi-task learning problem [15,16]. In general, classification tasks aim to learn the intrinsic semantic information of objects, while localization tasks derive the spatial locations of objects [17,18]. However, the performance of the two key tasks are not consistent, which undermines the overall detection accuracy. It is worth noting that most detection methods generate classification and location information at the same time, which is particularly prominent in the single stage framework [19]. Therefore, how to design an effective prediction mechanism is particularly necessary.

In recent years, considerable progress has been made in the single-stage object detection method due to its excellent real-time performance and accuracy, and the state-of-the-art (SOTA) detectors have demonstrated promising performance in many downstream applications [20–23].

However, these detectors have not taken into account the fundamental differences and relationship between localization and classification tasks. As a result, the prediction head is simply designed to generate a set of universal features required by the two subtasks at the same time, which limits their high-quality (intersection of union, $\mathrm{IoU} > 0.5$) detection accuracy [1,26,27]. As shown in Fig. 1, the test results of existing object detectors on public datasets show that there is inconsistency between object localization and classification. The main reason is that the coupled heads have difficulty making accurate and consistent predictions for both tasks. Thus, the existing prediction

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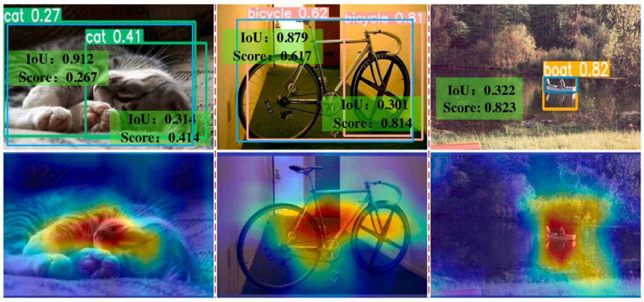

Fig. 1. The related example visualizations using YOLO [24]. The upper side shows some examples containing classification score and IoU, and blue boxes represent ground-truth. The lower side shows the Grad-CAM response area [25]. The current methods fail to achieve consistent prediction of localization and classification for some targets.

networks have deficiencies, which limit the detector to obtain accurate bounding boxes and thus undermine the overall accuracy.

At present, the research field have recognized this problem and proposed a series of parallel branches and task-aligned heads to further improve the detection accuracy [28–33]. Unfortunately, despite the efforts, these methods still have two main limitations, which have significantly reduced the accuracy in object detection.

(1) Feature differences between classification and location. Some researches replace the shared convolution layer (coupling head) by designing parallel predictive branches, which does not fully recognize the difference of task specific characteristics [4,14,34–36]. That is, the classification task emphasizes on the importance of channels, while the regression task pays more attention to the spatial information of images. Therefore, this decoupled method destroys the performance of the two subtasks to a certain extent.

(2) Correlation of classification and location tasks. Most of the existing decoupled heads mainly focus on the independence of regression and classification tasks, and ignore the correlation between the prediction results of the two subtasks, resulting in difficulties in obtaining high-quality detection results [37–40]. In fact, in actual object detection, an excellent bounding box (detection result) should have both high classification confidence (classification prediction) and location score (location prediction).

The development of the attention mechanism provides a new idea for this problem. The classification and location tasks can be better optimized by establishing effective attention blocks. Some studies [41–46] show the strong advantages of channel attention and spatial attention module in the backbone network for various challenging visual fields. The decoupled detection head tends to have a more reliable processing paradigm from this point of view. On the one hand, leveraging these insights, we can extract spatial information and semantic channel to better achieve both localization and classification tasks. On the other hand, we establish a connection and reinforcement mechanism of multi task features to promote the consistency of classification and regression predictions.

Based on the above analysis, we propose a task-guided attention-decoupled head, which includes a task decoupled attention distributor and a task correlation network. The former completes accurate localization and classification by designing Multi-Scale Semantic Intensifier (MSSI) and Large-Kernel Spatial Perceptron (LKSP). The latter establishes a correlation connection mechanism for different subtasks to make consistent predictions. The proposed approach is lightweight and can be readily applied in combination with any detectors to improve their overall performance. On the MS-COCO test-dev, we have completed 53.5 AP, which is 1.9 AP higher than the SOTA methods on average.

The contributions are summarized as follows:

(1) We propose an effective Task-Guided Attention Decoupled Head (TGADHead), which can be embedded into any single-stage detector as a prediction network, greatly improving the classification and regression performance in object detection tasks.

(2) We propose a Task Decoupled Attention Distributor (TDAD), which allocates task specific attention weights to enhance semantic and spatial information, thereby avoiding the misplacement of classification and localization features.

(3) In order to better transmit the information between classification and regression, we introduce a Task Correlation Network (TCN) to combine and strengthen both features for overall accurate prediction.

2. Related work

Single-stage Object Detectors. Object detection is a critical and challenging task in computer vision and the basis for many advanced tasks [47,48]. In particular, single-stage detectors have been used in research and industry [49]. As the most widely applied YOLO series, its simplified network architecture and powerful prediction mechanism have brought significant progress to the object detection. Subsequently, [50] proposed Focal Loss to focus on the problem of imbalanced sample classification and further build a powerful single-stage detector RetinaNet. As far as regression task is concerned, some studies [21,51] designed boundary enhancement modules with coarse-to-fine precise features to improve object detection accuracy. Based on the anchor-free mechanism, FCOS [34] improved the prediction performance of classification and regression subtasks through prediction decoupling. RefineDet [52] committed to obtaining accurate bounding boxes through the negative anchor screening and refinement mechanism. Based on the knowledge distillation strategy, [53] migrated the knowledge obtained from the main network to the downstream target network to achieve dense target detection. Although the above methods have achieved landmark results in many aspects, the prediction head is still a necessary structure that is often ignored. More importantly, existing prediction networks infer the attributes of objects from the same feature map grid, which may result in degradation to location and classification results. Therefore, how to design a reasonable prediction mechanism is still the key problem.

Detection Head. As a key component for object detection, detection head is designed to predict object categories and bounding box locations [57]. In the prediction mechanism of single-stage object detection, the results of classification and regression need to be generated at the same time. As shown in Fig. 2(a), in YOLO (v3-v5), each pixel of the features predicts the location and category of the anchor point, and finally derives the fine detection through Non-Maximum Suppression (NMS) [58]. In the RetinaNet, a parallel non-shared convolution branches is designed for localization and classification information prediction. As shown in Fig. 2(b), this parallel branch does not fully consider the characteristics of the tasks, i.e., the spatial and semantic information of the features are ignored. Further, in some recent studies as shown in Fig. 2(c), task-aligned heads are designed to balance classification and regression tasks by designing adaptive weight assignments. In [59], a category specific attention network was proposed to achieve adaptive category feature refinement and improve the recognition accuracy of downstream tasks. For example, in [60], a mutual learning framework with task alignment combined with ensemble learning ideas was proposed to greatly improve the detection accuracy of diverse targets. In [61], an attention guided task alignment head was designed to enhance classification and localization features, promote interaction and alignment between the two, and improve detection accuracy. Although the task-aligned structure improves the performance of existing prediction mechanisms, this type of method using a single weight layer to simultaneously align classification features and localization features is not conducive to fully mining the key information to improve

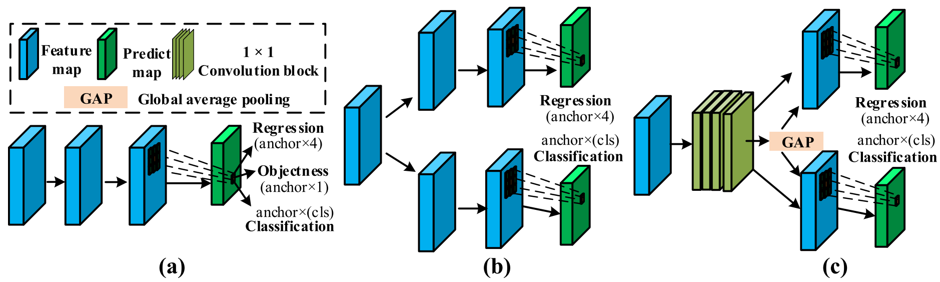

Fig. 2. Existing detection head pipeline. (a): YOLO(v3-v5) series of coupled heads (shared convolution layer). (b) RetinaNet [50], ATSS [54] series of decoupled heads. Two task features are extracted through two parallel branches [55]. (c) Task-aligned heads [32,56]. A single weight layer is designed to assign two specific tasks at the same time, which limits the overall detection performance.

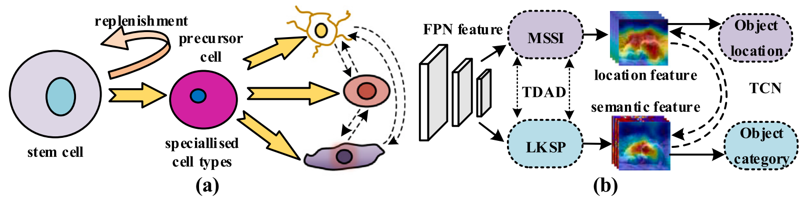

Fig. 3. Stem cell differentiation process (a) and simplified TGADHead framework (b). In TGAD, we capture task specific key representations with discriminative properties through TDAD and coordinate correlations through TCN. This process is similar to the idea of cell differentiation that separates first and then coordinates, and encourages us to use a divide and conquer approach to complete the target prediction task.

the performance of two subtasks [32,62]. In addition, the predictive relevance and consistency between tasks have not been fully explored.

Attention Mechanism. Inspired by the human visual system, the attention mechanism can selectively strengthen the region of interest and suppress complex background noise. In [42], in order to strengthen the feature representation ability of objects, a coordinate attention is further proposed on the basis of SENet [43] to improve the performance of detection and segmentation tasks. In [44], a dynamic convolution mechanism based on spatial transformer is proposed to effectively balance accuracy and efficiency. In [59], a category specific attention network is proposed to achieve adaptive category feature refinement and improve the recognition accuracy of downstream tasks. In our framework, inspired by the idea of divide and conquer, we distribute task-specific attention for classification and location features to enhance the multi-task learning ability of the detector.

3. Task-guided attention-decoupled head

3.1. Approach overview

Motivation. In order to generate predictive features specific to classification and bounding box regression tasks, we took inspiration from cell biology [63,64]. As shown in Fig. 3(a), during the differentiation of biological stem cells, embryonic stem cells can differentiate into cells or tissues with different functions and properties under the guidance of specific exogenous inducers. Inspired by stem cell differentiation, the structure of the proposed simplified version is shown in Fig. 3(b). Similarly, FPN generates enhanced features suitable for classification and localization, which are interdependent and jointly make the final prediction.

Overview. In this work, we propose a TGADHead for single-stage detector to achieve high accuracy object detection tasks (as shown in Fig. 4). First, given an image $\mathcal{X}$, a series of multi-level semantic and spatial features are generated through a backbone network Φ and a neck network $\mathcal{F}$, i.e., $\{F_i\}_{i=1}^{n} = \mathcal{F}(\Phi(\mathcal{X}))$, where $F_i \in \mathbb{R}^{H_i \times W_i \times C_i}$. After this, the $\{F_i\}_{i=1}^{n}$ are input to the proposed TGADHead to complete the localization and classification branches. In the detection head, the multi-scale semantic intensifier $\mathcal{I}_{ms}$ and the large-kernel spatial perceptron $\mathcal{P}_{ls}$ are designed to decouple the classification task and the regression task to alleviate feature misalignment. Furthermore, the task correlation network $\mathcal{N}_{tc}$ is presented to transfer the correlation information between the features required for classification and regression tasks, so as to facilitate them to make accurate predictions. Finally, multi-task learning loss $\mathscr{L}_{det}$ is constructed to optimize all network parameters.

3.2. Task decoupled attention distributor

The focus of this work is to build a more effective task-decoupled attention mechanism. As shown in Fig. 4, the TDAD mainly includes two attention blocks. On the one hand, it strengthens the semantic information between channels to improve the classification performance, and on the other hand, it improves the spatial information to improve the localization accuracy. For better feature aggregation, we adopt 1×1 convolution to aggregate FPN features F_i to $F_m \in \mathbb{R}^{C \times H \times W}$. Then the two unshared convolutions are used to compute the classification branch features $F_c \in \mathbb{R}^{C \times H \times W}$ and regression branch features $F_s \in \mathbb{R}^{C \times H \times W}$.

Multi-scale Semantic Intensifier. Structurally, MSSI is inspired by the SENet [43] to enhance the interaction between feature channels in the network output, thereby improving the classification accuracy. MSSI is divided into two steps: multi-scale dimension squeezing and channel interaction, which are respectively used for global information embedding and adaptive reweighting of channel relationships. In order to reduce computing resources, group convolution is applied to calculate the generated features of different kernel scales, i.e.,

$$S_i^g = Conv_{k_i \times k_i, g_i}^{C/4}(Conv_{1\times 1}^{C/8}(F_c)), i \in \{0, 1, 2, 3\} \tag{1}$$

where $Conv_{k\times k}^{c}$ represents the $k \times k$ convolution, and the number of channels is c, the ith kernel size $k_i = 2i + 3$, the ith group size $g_i = 2^{(k_i-1)/2} \in \{1, 4, 8, 16\}$, $S_i^g \in R^{W \times H \times C/4}$ denotes the feature map with different scales. Then, the information-rich features can be obtained, i.e.,

$$S_c^g = Concat([S_0^g, S_1^g, S_2^g, S_3^g]) \tag{2}$$

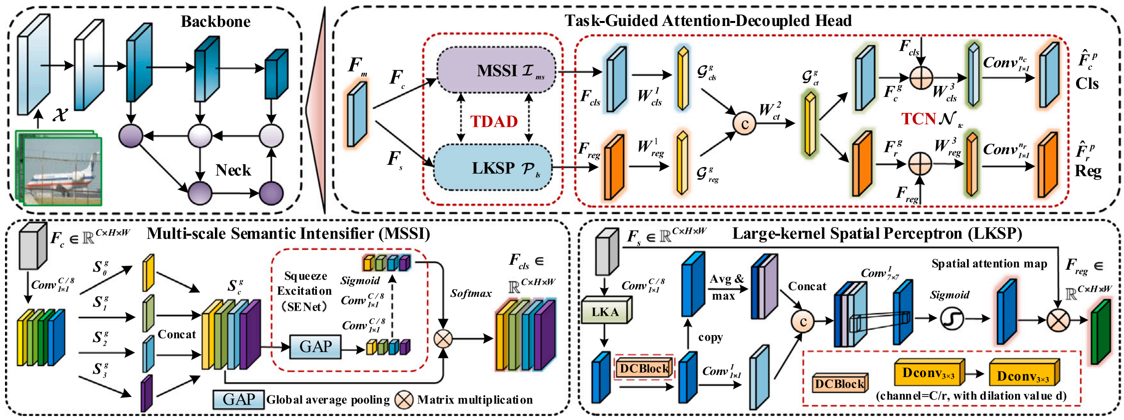

Fig. 4. Single-stage object detection framework with task-guided attention-decoupled head. Overall detection network consists of a backbone (Modified CSPNet), a neck (PAN) and a head (TGADhead). In the detection head, we first proposed a Task Decoupled Attention Distributor (TDAD) to separate the classification and regression tasks. In particular, for classification, we designed a multi-scale semantic intensifier (MSSI) to enhance the importance of relevant channels, while for regression task, we designed a large-kernel spatial perceptron (LKSP) based on large kernel attention (LKA) [41] to enhance spatial features. Finally, the task correlation network (TCN) is constructed to obtain the final task-specific prediction map $\hat{F}_c^p \in \mathbb{R}^{H_i \times W_i \times n_c}$, $\hat{F}_r^p \in \mathbb{R}^{H_i \times W_i \times n_r}$.

Next, we integrate the multi-scale features S_c^g through squeezing and excitation operations (SE-Net).

$$\omega_i^g = \sigma(T_1(T_2(\frac{1}{H \times W}\sum_{h=1}^{H}\sum_{w=1}^{W} x_c^g(i,j)))) \tag{3}$$

where $T_1(\cdot)$ and $T_2(\cdot)$ represent convolution operations for performing squeeze-and-excitation process, $x_c^g(i,j) \in S_i^g$.

The multi channel weight $\omega^g = Concat(\omega_0^g, \ldots, \omega_3^g) \in \mathbb{R}^{C\times 1\times 1}$, which includes information from all channels. To capture the correlation between channels, we perform $Softmax$ activation on the weighted ω^g after excitation, i.e.,

$$\hat{att}^c = \frac{exp(\omega_i^g)}{\sum_{i=1}^{C} exp(\omega_i^g)} \in \mathbb{R}^{C\times 1\times 1} \tag{4}$$

where ω_i^g represents the ith scale channel weight. Based on residual connection, we obtain the prediction features F_{cls} suitable for classification tasks, ie. $F_{cls} = F_c \otimes \hat{att}^c \oplus F_c$, where $\otimes$ refers to the element-wise multiplication.

By squeezing and exchanging information on F_c, we obtain the attention weight $\hat{att}^c$ between different scale channels. Those feature channels F_{cls} with strong correlations are implemented to generate the classification probability.

Large-kernel Spatial Perceptron. We utilize the LKSP to establish the long-distance dependent information in space, and then further explore the deep receptive field, focusing on where the object is. Then, the attention map $\hat{att}^s$ carrying targeted spatial location information is recombined with the F_s to obtain regression features F_{reg}.

Specifically, the feature $F_s \in \mathbb{R}^{C\times H\times W}$ is mapped $\tilde{F}_s \in \mathbb{R}^{C/r\times H\times W}$ to compress and integrate channel information. Then, inspired by the visual attention network, the large kernel convolution (e.g., $k = 21$) is decomposed into deep-wise convolution $DWConv$, depth-wise dilation convolution $DWDConv$ and standard convolution $Conv$, which can effectively reduce the number of parameters and establish the long-distance spatial dependence [65,66].

$$\psi = Conv_{1\times 1}^{C/r}(DWConv_{5\times 5}^{C/r}(DWDConv_{7\times 7}^{C/r}(\tilde{F}_s))) \tag{5}$$

where $\psi \in \mathbb{R}^{C/r\times H\times W}$ refers to the large kernel attention used to establish long-distance dependence, and then obtain the primary feature $\hat{F}_s \in \mathbb{R}^{C/r\times H\times W}$, i.e., $\hat{F}_s = \psi \otimes \tilde{F}_s$.

In the bounding box regression task, improving the receptive field is helpful to obtain more accurate target location. Therefore, we design a dilated convolution block $DCBlock$ to further strengthen spatial information and effectively utilize global context. For simplicity, we fix the channel dimension of the guidance feature F_g^s, which are calculated as follows:

$$F_g^s = Swish(DCBlock(Conv_{1\times 1}(\hat{F}_s))) \tag{6}$$

where $DCBlock$ contains two layers 3×3 dilated convolutions ($r = 16, d = 4$) and is designed to focus global context information with large receptive fields.

To calculate the spatial attention weight $\hat{att}^s$, we further apply average pooling $AvgPool(\cdot)$, maximum pooling $MaxPool(\cdot)$ and convolution operations $Conv_{1\times 1}(\cdot)$ along the channel axis of guide feature F_g^s to generate three representative features $[F_{avg}; F_{max}; F_{conv}]$, which are concatenated to form the input features for subsequent convolution operations. Thus, the process can be expressed as:

$$\begin{cases} F_{avg} = AvgPool(F_g^s) \\ F_{max} = MaxPool(F_g^s) \\ F_{conv} = Conv_{1\times 1}(F_g^s) \end{cases} \tag{7}$$

where $F_{avg}, F_{max}, F_{conv} \in \mathbb{R}^{1\times H\times W}$ represents the single channel attention map with rich spatial information extracted from different angles. Finally, the output of the large-scale convolution operation will be activated by the $Sigmoid$ to obtain the $\hat{att}^s$, i.e.,

$$\hat{att}^s = \sigma(Conv_{7\times 7}[F_{avg}; F_{max}; F_{conv}]) \in \mathbb{R}^{1\times H\times W} \tag{8}$$

where $\hat{att}^s$ represents the spatial attention map. Similarly, we use matrix multiplication and residual connection to obtain F_{reg}. That is, $F_{reg} = F_s \otimes \hat{att}^s \oplus F_s$.

In the above operations, we obtain the spatial perception map in two steps. The former obtains sufficient large-scale spatial information by focusing on the long-distance dependence, and the latter strengthens the object location feature by mining the deep receptive field, thus providing powerful prediction features for the bounding box regression task.

3.3. Task correlation network

In order to promote the consistency of classification and regression task prediction, inspired by the graph network [67,68], we construct learnable correlation network for multi-task features $\{F_{cls}, F_{reg}\}$, and strengthen the association between different tasks through the learning correlation node parameters. The TCN consists of three 1D convolutional layers ($\{W_{cls}^1, W_{reg}^1\}, W_{ct}^2, \{W_{cls}^3, W_{reg}^3\}$). In order to convert the features $\{F_{cls}, F_{reg}\}$ into higher semantics to obtain sufficient representation ability, we constructed the following transformation, i.e.,

$$\begin{cases} \mathcal{G}^g_{cls} = W^1_{cls}\mathcal{T}_1(F_{cls}) \\ \mathcal{G}^g_{reg} = W^1_{reg}\mathcal{T}_1(F_{reg}) \end{cases} \tag{9}$$

where $\mathcal{T}_1$ means to convert the spatial dimension $\mathbb{R}^{C\times H\times W}$ to $\mathbb{R}^{C\times(HW)}$. Then, we use interactive operations to perform tensor integration on multi-task features $\{\mathcal{G}^g_{cls}, \mathcal{G}^g_{reg}\}$, $\mathcal{G}^g_{ct} = W^2_{ct}\mathcal{T}_1[\mathcal{G}^g_{cls} \parallel \mathcal{G}^g_{reg}]$, where $\parallel$ represents tensor connection.

Now that we have obtained the interaction feature $\mathcal{G}^g_{ct}$, the operation of split and conquer will be performed, i.e.,

$$\begin{cases} F^g_c = W^3_{cls}(\mathcal{T}_1(\mathcal{T}_2(\mathcal{G}^g_{ct}) + F_{cls})) \\ F^g_r = W^3_{reg}(\mathcal{T}_1(\mathcal{T}_2(\mathcal{G}^g_{ct}) + F_{reg})) \end{cases} \tag{10}$$

where $\mathcal{T}_2$ represents the inverse operation of $\mathcal{T}_1$.

Next, we employ 1×1 convolution function to perform prediction, the final prediction map of each task is:

$$\begin{cases} \hat{F}^p_c = Conv^{n_c}_{1\times1}(\mathcal{T}_1(F^g_c)) \\ \hat{F}^p_r = Conv^{n_r}_{1\times1}(\mathcal{T}_1(F^g_r)) \end{cases} \tag{11}$$

where $\hat{F}^p_c$, $\hat{F}^p_r$ represent the classification map, regression map of the prediction, n_c, n_r represent the number of categories and the maximum number of regression.

By building the TCN, the dependencies between tasks are established, which is used to accelerate convergence and obtain efficient inference performance for the model.

3.4. Model training

In this section, we establish the multi task learning loss $\mathscr{L}_{det}$ based on the $\{\hat{F}^p_c, \hat{F}^p_r\}$. It consists of two components: (1) DFL-based bounding box regression loss $\mathcal{L}_{reg}$ [69,70]; (2) the loss for classification of objects $\mathcal{L}_{cls}$. Compared with $\mathcal{L}_{reg}$, the latter uses TOOD label allocation [32] and binary cross entropy. The $\mathscr{L}_{det}$ can be formulated as:

$$\begin{aligned} \mathscr{L}_{det} = \lambda_r \mathcal{L}_{reg} + \lambda_c \mathcal{L}_{cls} = \sum_{i=1}^{N_p} \lambda_r L_{CIoU}(b^p_i, b^t_i) + \\ \lambda_c \left(\sum_{i=1}^{N_p} \xi_p L_B(p^p_i, c^t_i) + \sum_{j=1}^{N_n} \xi_n L_B(p^p_j, 0) \right) \end{aligned} \tag{12}$$

where λ_r and λ_c represent the hyper-parameters that control the $\mathcal{L}_{reg}$ and $\mathcal{L}_{cls}$ respectively. (N_p, N_n) and (ξ_p, ξ_n) represent the number of positive and negative samples and the degree of training, respectively, L_B denotes the binary cross-entropy function, expressed by,

$$L_B(p^p_i, c^t_i) = -[p^t_i \cdot \log(p^p_i) + (1 - p^t_i)\log(1 - p^p_i)] \tag{13}$$

where p^p_i and c^t_i represent the probability and positive label under anchor ith, b^p_i and b^t_i denote the predicted bounding boxes and the corresponding ground-truth boxes. In the inference process, we decode the regression prediction map $\hat{F}^p_r$ based on DFL to obtain the boundary box of the object with confidence.

4. Experimental results

4.1. Experimental settings

Datasets: The large-scale MS-COCO2017 with 80 categories and the medium-sized Pascal VOC 2007 with 20 categories are used for experimental establishment. For the VOC 2007, we train on trainval with 5K images and report test results with 5K images. We conduct an ablation study to analyze the effectiveness of different components in the proposed framework. For the MS COCO2017, we train on train2017 (118K images) and report the results on val2017 (5K images) and test-dev (41K images) for comparison and analysis with state-of-the-art methods.

Evaluation Metric: The standard COCO dataset style AP with different IoU thresholds from 0.5 to 0.95 is used as evaluation metric. Further, AP_l, AP_m, and AP_s are used to evaluate the large ($area > 96^2$), medium ($32^2 < area < 96^2$), and small ($area < 32^2$) object detection performance of the target bounding box, respectively. Similarly, for the Pascal VOC2007 dataset, we calculate $AP_{0.5}$ and AP.

4.2. Implementation details

Architecture: Our TGADHead is evaluated in detail on four backbones of different widths and depths, all trained from scratch and initialized randomly. For a fair comparison, we adopt the novel modules in [24] to construct the backbone, mainly including the modified CSPNet and PAN. We also generate four groups of modified models according to their scaling rules, i.e., {CSP-s, CSP-m, CSP-l, CSP-x}.

Hyper-parameters: The initial learning rate is set to 0.01 and 3 epoch warm-up is performed. We use the standard SGD optimizer with decay and momentum of 0.9 to train all the models. The weight decay is set to 0.0005. We use four NVIDIA GPUs for training, set the batch size to 64 and adopt BN layer optimization. All models are fine-tuned with 300 epochs. The loss coefficients λ_r and λ_c are set to 7.5 and 0.5, respectively.

Data augmentation: We mainly use two violent scaling methods and image mosaic methods for data enhancement, and apply them to ablation study models with different sizes. The specific parameters are as follows: scaling probability $P_{scale} = 0.9$, $P_{flip} = 0.5$, $P_{mosaic} = 1$, $P_{mixup} = 0.1$. Following YOLOx's setting, we stopped using strong data augmentation in the last 30 rounds of model training. In addition, when performing scale enhancement, we follow standard settings, randomly increasing in a scaling form from $(1-P_{scale})$ to $(1+P_{scale})$, and filling the reduced area with 114 pixels. This violent scale enhancement is very beneficial for simulating real-world application scenarios.

4.3. Effectiveness analysis

4.3.1. Ablation studies

We conduct detailed ablation studies on two datasets to verify the effectiveness of each module. The specific implementations are as follows:

(a) Coupled head: The detection head shown in Fig. 2(a) is applied to perform classification and regression prediction.

(b) Decoupled head: We adopt a decoupled head by using two parallel branches (Remove TDAD and TCN).

(c) TDAD-based decoupled head: We employ MSSI and LKSP to achieve classification and regression prediction.

(d) TGADHead: TDAD-based decoupled head+TCN (see Table 1).

The decoupled detection head is designed to capture task-related features and guide the result prediction to improve the model's predictive power for different targets. Compared to using only the coupled head, TDAD brings improvements of **3.8%**, **4.1%**, **2.4%**, and **1.8%** on four backbones of different sizes, respectively. More specifically, it can be seen from Table 3 that the performance on the VOC2007 dataset can be improved again by introducing attention mechanism. Further, task correlation network networks help multiple tasks cooperate with each other to achieve the best prediction and ultimately the highest detection accuracy. The TGADHead achieved the powerful results (**58.3** AP). In Table 2, we also report the detection results of MS-COCO2017 val (+**6.1%** AP compared to baseline), which implies that our method is equally competitive on large-scale datasets.

As shown in Fig. 5, we visualized the feature attention map generated by the four head models. It can be found that the detection head with task-decoupled attention module can better help the detector localize the object of interest compared with the coupled detection head and the conventional decoupling. We argue that the advantages of our proposed TDAD structure with location information and semantic boosting compared to the original baseline model are twofold. First, the

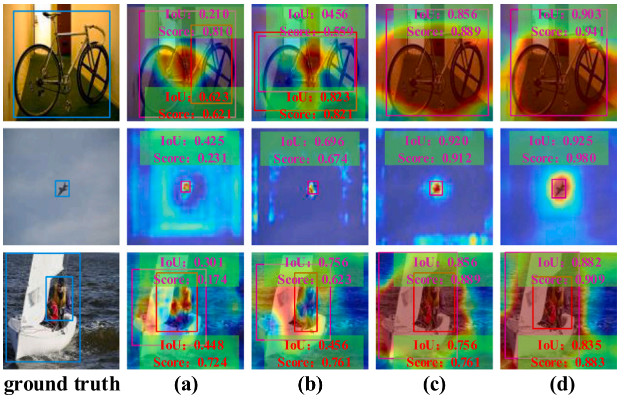

Fig. 5. Visualization of feature maps base on Grad-CAM. (a) (b) (c) (d) represent the coupled head, the decoupled head, the TDAD-based decoupled head and TGADHead respectively.

Table 1
Object detection results of Pascal VOC 2007 test set. The above prediction heads were inserted into four different backbone networks for comparative experiments.

No.	Backbone	$AP_{0.5}$	AP	P	R
(a)	Modified CSP-s	65.9	39.9	69.7	63.1
(b)	Modified CSP-s	67.7	41.8	70.3	64.5
(c)	Modified CSP-s	69.7	45.5	71.0	65.3
(d)	Modified CSP-s	**70.9**	**48.7**	**73.0**	**65.8**
(a)	Modified CSP-m	70.6	46.5	72.2	65.7
(b)	Modified CSP-m	71.8	48.1	73.5	67.3
(c)	Modified CSP-m	74.7	50.5	75.3	69.0
(d)	Modified CSP-m	**75.2**	**51.5**	**76.1**	**70.1**
(a)	Modified CSP-l	73.1	49.5	77.2	68.2
(b)	Modified CSP-l	73.7	50.7	78.9	67.4
(c)	Modified CSP-l	75.5	53.9	78.0	70.6
(d)	Modified CSP-l	**77.0**	**56.6**	**79.2**	**72.0**
(a)	Modified CSP-x	76.3	51.0	78.4	70.1
(b)	Modified CSP-x	77.0	53.0	78.7	70.6
(c)	Modified CSP-x	78.1	57.0	**80.5**	71.9
(d)	Modified CSP-x	**79.5**	**58.3**	**81.3**	**72.5**

Table 2
Speed and accuracy comparison on MS-COCO 2017 val, the adopted backbone is the modified CSP(s)-Net.

No.	AP	$AP_{0.5}$	$AP_{0.75}$	Parameters	GFLOPs
(a)	37.7	56.9	36.3	**7.2** M	**16.5**
(b)	38.6	58.3	40.4	7.9 M	18.7
(c)	41.4	59.5	44.7	9.3 M	22.5
(d)	**43.8**	**61.9**	**47.1**	11.0 M	26.5

multi-scale semantic intensifier improves the semantics of the target for higher confidence scores, and the large-kernel spatial perceptron helps to capture strong target edge information for localization. Secondly, by adding task correlation network, the proposed TGADHead obtains more accurate bounding boxes.

To further demonstrate the inherent advantages of task specific predictive decoupling mechanisms, we also conducted more extensive head to head ablation. We have set the following modes on the VOC test set using the Modified CSPs backbone network mentioned above. From the results, it can be observed that the proposed task specific attention decoupling mechanism has more advantages than single attention based mechanisms (see Table 3 for details). In addition, when using TCN alone, there is a certain improvement in the model's AP metric, indicating that it encourages high-quality localization and classification. However, due to the lack of TDAD mechanism, the localization and classification performance of $AP_{0.5}$ is slightly weak. Furthermore, from Table 4, it can be observed that introducing large kernel attention can increase the receptive field of the model localization branch, thereby promoting the detector to improve the accuracy of target recognition. When the large kernel scale is continuously increased, the performance of the model will have a slight improvement, and it will tend to stabilize when it reaches 21 * 21.

Table 3
More extensive effectiveness analysis for task specific attention distributors on pascal VOC 2007 test set.

Mode	$AP_{0.5}$	AP	P	R
MSSI(cls)+MSSI(cls)	68.9	44.1	69.1	64.0
LKSP(reg)+LKSP(reg)	69.2	44.3	68.3	64.7
LKSP(cls)+MSSI(reg)	69.0	42.7	70.8	65.1
MSSI(cls)+LKSP(reg)	**69.7**	45.5	**71.0**	**65.3**
Single TCN	67.8	**45.8**	69.8	60.1

Table 4
Analysis of convolutional kernel scale performance for large kernel attention on pascal VOC 2007 test set.

LKA mode	$AP_{0.5}$	AP	P	R
7*7 scale	69.7	48.3	72.5	64.6
14*14 scale	70.2	48.5	72.0	64.8
21*21 scale	**70.9**	48.7	73.0	**65.8**
28*28 scale	70.6	**49.0**	**73.2**	65.0

Table 5
Test results generated by adjusting TGADHead structure order based on Modified CSP-x backbone and PAN neck.

Method	Dataset	P	R	$AP_{0.5}$	AP
TCN+TDAD	VOC2007	78.0	71.5	77.3	53.2
TDAD+TCN	VOC2007	**78.7**	**72.0**	**77.8**	**55.4**
TCN+TDAD	COCO	70.5	59.5	66.5	49.9
TDAD+TCN	COCO	**74.4**	**61.6**	**70.1**	**53.5**

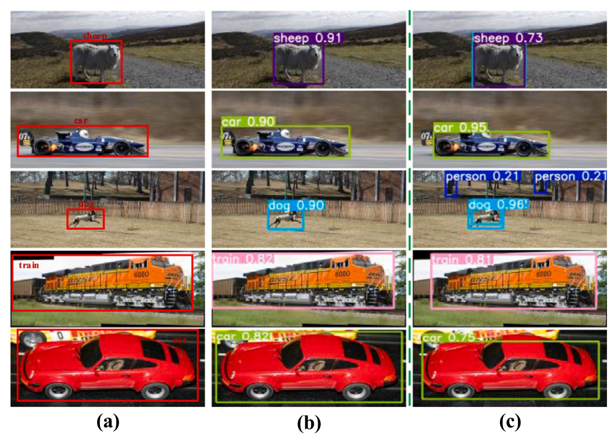

Fig. 6. The performance analysis under different network structure order is in VOC2007 test set. (a) Displays labels for related instances with target locations and categories. (b) and (c) represent the test results under TDAD+TCN (TGADHead) and TCN+TDAD methods respectively.

4.3.2. Structural analysis

In our TGADHead framework, the task-decoupled attention distributor (TDAD) is placed in front of the task correlation network (TCN). Therefore, we will conduct supplementary experiments and discussions on the module position order.

Quantitative results. The results in Table 5 show that FPN features are first input to TDAD, and then the relevant network is executed. The detection head structure can better achieve accurate object prediction. If the order of two key modules is adjusted, there will be a significant decline in performance, because the prediction mechanism tends to learn the characteristics of specific tasks first and then learn

Table 6
Visualization of detection results on the MS-COCO 2017 test-dev. TGADhead shows excellent detection performance.

Method	Backbone	Size	AP	$AP_{0.5}$	$AP_{0.75}$	AP_s	AP_m	AP_l
Faster R-CNN w/ SABL [71]	ResNet-101	800	41.8	60.2	45.0	23.7	45.3	52.7
Cascade R-CNN [72]	ResNet-101	800	42.4	61.1	46.1	23.6	45.4	54.1
FCOS [34]	ResNet-101-FPN	800	41.5	60.7	45.0	24.4	44.8	51.6
BorderDet [73]	ResNet-101-FPN	800	43.2	62.1	46.7	24.4	46.3	54.9
Double-Head-Ext [74]	ResNet-101-FPN	800	42.3	62.8	46.3	23.9	44.9	54.3
RetinaNet [50]	ResNeXt-101-FPN	640	40.8	61.1	44.1	24.1	44.2	51.2
FSAF [75]	ResNeXt-101-FPN	640	42.9	63.8	46.3	26.6	46.2	52.7
ATSS [54]	ResNeXt-101-FPN	640	47.7	66.5	51.9	29.7	50.8	59.4
TOOD [32]	ResNeXt-101-FPN	640	49.3	68.6	53.5	31.2	53.2	62.0
VarifocalNet [57]	ResNeXt-101-FPN	640	46.1	64.7	50.4	27.8	49.8	56.7
TSD [76]	ResNeXt-101-FPN	640	50.0	68.9	54.0	32.8	53.8	63.2
w/G-RFA [33]	ResNeXt-101-FPN	640	49.3	68.3	53.6	30.3	52.4	61.4
DCFA [77]	ResNeXt-101-FPN	640	50.8	**70.0**	56.2	32.5	55.0	64.6
TGADHead (Ours)	ResNeXt-101-FPN	640	**51.9**	69.3	**56.7**	32.2	**56.3**	**65.2**
YOLOv5 [25]	Modified CSP-x-PAN	640	49.5	67.7	56.5	32.7	55.2	64.9
YOLOv7 [78]	Modified CSP-x-PAN	640	52.7	**70.4**	56.9	33.5	56.3	65.8
TGADHead (Ours)	Modified CSP-x-PAN	640	**53.5**	70.1	**57.7**	**34.2**	**56.8**	**66.2**


(a) (b) (c)

Fig. 7. The performance analysis under different network structure order is in COCO val2017. Similar (a) represents the original label, (b) and (c) represent the test results under TGADHead and TCN+TDAD methods respectively.

the semantics of relevance. Therefore, TGADHead by the above analysis is reasonable and effective.

Result visualization. We compare the detection results of the two structures and obtain the visual analysis as shown in Figs. 6 and 7. By comparing the detection information with the ground truth, it can be found that TGADHed structure has stronger detection capability and can accurately detect dense objects, small objects and large objects.

Heat map visualization. We have made a visual analysis of the heat maps of the two different structures. As shown in Figs. 8 and 9, TGADHead can effectively learn the spatial information and semantics of objects, and focus reasonable attention on task-specific perception. Therefore, the proposed TGADHead can be used as an effective prediction mechanism as a plug-and-play single-stage object detection head and further improve the overall object detection accuracy.

4.4. Comparison with the SOTA method

4.4.1. Comparison with other object detection methods

In this section, we show a detailed comparison between the proposed method and the state-of-the-art methods (including single-stage and two-stage detectors, anchor-based and anchor-free detectors) on the MS COCO test-dev dataset. As shown in Table 6, our method achieves the highest accuracy (**53.5** AP) without any additional tricks, which is obviously superior to other mainstream methods, including the Cascade R-CNN (42.4 AP), the FCOS (41.5 AP), the ATSS (47.7 AP), the YOLOv5-CSP (49.5 AP), the YOLOv7-CSP (52.7 AP). Furthermore, for fairness, we also analyze some advanced methods of decoupled detector. It can be found that compared with TSD, TOOD and DCFA of the proposed method is improved by **2.0** AP, **2.7** AP and **1.1** AP respectively. The visualization results are shown in Fig. 10.

Compared with other methods, TGADHead has advantages in performance for two main reasons. First, the proposed TDAD is a task specific attention mechanism, which can improve the prediction characteristics of bounding box regression tasks and classification tasks. Furthermore, we set up a task correlation network for the two tasks to avoid the complete independence and separation of the two subtasks. Therefore, these two tasks can coordinate with each other to make accurate and consistent predictions.

4.4.2. Comparison with other prediction mechanism methods

Performance Analysis. In this section, we have discussed more about the performance of the proposed TGADHead. We supplemented and analyzed some mainstream decoupling methods, including YOLOX decoupling head [55], PPYOLOE head [62], TOOD [32], YOLOv6-Head [56], YOLOv7-Head [78], etc. Table 7 shows the performance of the COCO dataset. We obtained competitive results without increasing the reasoning time. We completed SOTA performance on stricter IoU threshold indicators, e.g., AP, $AP_{0.75}$, and different scale indicators. Compared to the standard TOOD and YOLOv7 detection heads, we achieved improvements of 1.2% and 0.9% in fair comparison. Similarly, Tables 8 and 9 show the performance comparison of VOC2007 dataset in the two modes (load COCO pre-training weights). It can be concluded that our method has improved performance in various categories of VOC2007 dataset. These sufficient results demonstrate that our task specific decoupling module is necessary.

Efficiency Analysis. Table 10 shows the operation efficiency comparison of various methods under the same backbone network. [55] uses a series of 3×3 convolution to perform classification and regression decoupling operations, which leads to significant increase in prediction time delay and parameters. [62] retains part of 3×3 convolution, and uses global average pooled adaptive classification and regression tasks. [56] performs task assignment operations with GAP through a series of 1×1 convolutions. Our method avoids the reasoning time delay caused by 3×3 convolution and improves the accuracy by using task decoupled attention distributor (TDAD) and task-related network (TCN).

Visualization Analysis. We have made a more detailed visualization of the detection results of TGADHead in VOC dataset and COCO dataset. Fig. 11 shows the detection effect of multiple groups of objects. By comparing Fig. 11(a) and (b), we can find that the classification confidence and location accuracy have not changed significantly after using our TGADHead to load the COCO pre-training model, which shows

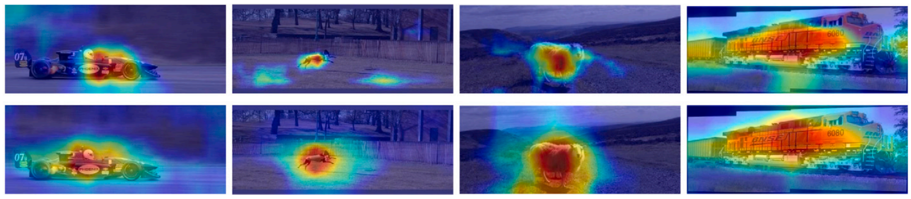

Fig. 8. Heat map visualization in VOC2007. The first line shows the results under TCN+TDAD method, and the second line shows the proposed TGADHead.

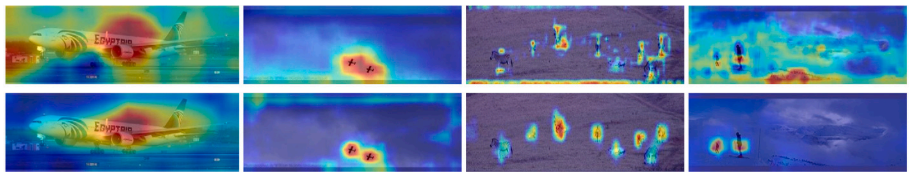

Fig. 9. Heat map visualization in COCO val2017. The first line shows the results under TCN+TDAD, and the second line shows the proposed TGADHead.

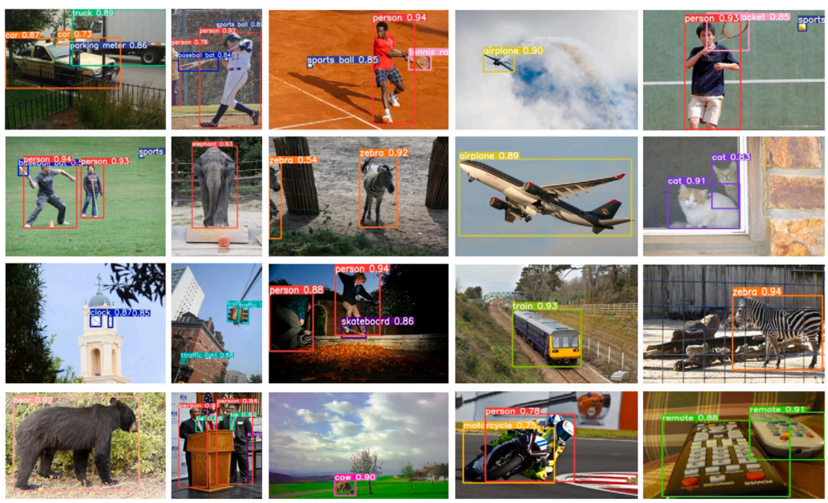

Fig. 10. More visualization of detection results on the COCO test2017. TGADhead shows excellent detection performance. From these representative example results, it can be found that targets of different scales can be accurately located and classified.

Table 7
Comparison of the speed and accuracy of different object detectors on COCO val 2017. Similarly, we use the modified CSP-s as the backbone network.

Method	AP	$AP_{0.5}$	$AP_{0.75}$	AP_s	AP_m	AP_l
YOLOx-Head [55]	38.2	57.1	36.7	22.1	42.0	48.1
PPYOLOE-Head [62]	41.8	60.2	45.0	23.7	45.3	52.7
TOOD-Head [32]	42.6	61.7	44.6	24.3	46.5	56.1
YOLOv6-Head [56]	42.4	61.5	45.2	25.7	46.8	55.6
YOLOv7-Head [78]	42.9	**62.1**	46.6	26.1	48.0	56.2
TGADHead(Ours)	**43.8**	61.9	**47.1**	**27.5**	**48.6**	**56.8**

that our method has task-specific prediction stability from another perspective. Fig. 11(c) shows the results of COCO val2017. The proposed method can obtain accurate object boundaries and high class prediction accuracy.

4.5. Analysis and discussion

In this section, we will discuss the performance of TGADHead under different validation methods based on the COCO val dataset.

4.5.1. Performance under different criteria

(1) **Performance in different IoU criteria**: In order to further analyze the performance of TGADHead in bounding box regression, we conducted a series of evaluations on COCO val2017 using the stricter IoU standard. Fig. 12(a) shows the performance comparison based on TGADHead and YOLO series head baseline (using the same backbone network). We can conclude that with the constant increase of IoU threshold (from 0.5 to 0.9), TGADHead performance declines more slowly than the baseline. This indicates that the proposed method has more high-quality positioning boxes.

(2) **Performance in different confidence criteria**: In target detection, confidence plays an important role in balancing the ratio of true positives and false positives. Generally speaking, the higher the confidence is, the higher the accuracy of target detection is. As shown in Fig. 12(b), with the increase of confidence threshold, our method achieves higher accuracy than the baseline method.

(3) **Performance in different scales criteria**: The effectiveness of TGADHead under different IoU standard and confidence standard is analyzed above. In order to better explore specific improvements, we further evaluated objects of different scales. Table 11 shows that TGADHead is successful in objects of different scales. It is noteworthy that with the increase of IoU threshold, the performance benefits brought by TGADHead are also improving. Finally, the visualization of the detection results of COCO val2017 is shown in Fig. 13. The proposed method can accurately identify the object categories and obtain high-quality regression boxes.

Table 8
More detailed method comparison for different detection heads on VOC2007 dataset. We have shown in detail the AP comparison of these methods in different categories, and our methods can achieve stable performance improvement in each category.

Method	Aero	Cycle	Bird	Boat	Bottle	Bus	Car	Cat	Chair	Cow	Table	Dog	Horse	Bike	Person	Plant	Sheep	Sofa	Train	Tv	map
YOLOx-Head [55]	0.796	0.809	0.554	0.483	0.456	0.749	0.855	0.673	0.483	0.696	0.594	0.627	0.809	0.767	0.804	0.380	0.627	0.578	0.761	0.677	0.659
PPYOLOE-Head [62]	0.804	0.804	0.572	0.532	0.496	0.744	0.863	0.693	0.508	0.709	0.630	0.662	0.829	0.776	0.812	0.432	0.639	0.604	0.758	0.688	0.677
TOOD-Head [32]	**0.812**	0.817	0.606	0.535	0.523	0.819	0.874	**0.767**	0.467	0.623	0.647	0.686	0.820	**0.794**	0.814	0.354	0.596	0.644	0.801	0.683	0.684
YOLOv6-Head [56]	0.770	0.807	0.612	0.536	**0.565**	0.775	0.869	0.698	0.520	0.692	0.615	0.654	0.804	0.783	0.800	**0.461**	**0.690**	0.583	0.755	0.646	0.683
YOLOv7-Head [78]	0.806	0.818	0.632	0.582	0.544	0.811	0.880	0.662	0.512	0.685	0.673	0.700	**0.832**	**0.794**	**0.848**	0.413	0.634	0.632	**0.822**	**0.700**	0.703
TGADHead(Ours)	0.790	**0.828**	**0.628**	**0.592**	0.514	**0.827**	**0.890**	0.754	**0.522**	**0.706**	**0.697**	**0.716**	0.827	**0.794**	0.833	0.422	0.678	**0.668**	0.798	0.686	**0.709**

Table 9
Test results of each category in VOC2007 dataset with pre-training weight of COCO data set.

Method	Aero	Cycle	Bird	Boat	Bottle	Bus	Car	Cat	Chair	Cow	Table	Dog	Horse	Bike	Person	Plant	Sheep	Sofa	Train	Tv	Map
YOLOx-Head [55]	0.872	0.864	0.713	0.619	0.658	0.846	0.904	0.781	0.584	0.778	0.647	0.760	0.861	0.827	0.853	0.503	0.741	0.664	0.817	0.744	0.752
PPYOLOE-Head [62]	0.867	0.874	0.710	0.608	0.664	0.862	**0.921**	0.792	0.597	0.753	0.716	0.790	0.887	0.872	0.868	0.513	0.740	**0.823**	0.849	0.734	0.767
TOOD-Head [32]	0.868	0.877	0.703	0.660	0.605	0.854	0.910	**0.821**	0.579	0.746	0.719	0.796	0.864	0.857	**0.873**	0.493	0.605	0.702	**0.856**	**0.763**	0.758
YOLOv6-Head [56]	0.882	0.867	0.727	0.624	0.676	0.834	0.903	0.813	0.622	**0.824**	0.655	0.792	0.885	0.734	0.868	0.530	0.764	0.693	0.833	0.759	0.770
YOLOv7-Head [78]	0.873	**0.890**	0.730	**0.685**	0.660	0.871	0.912	0.812	0.615	0.789	**0.723**	0.788	**0.901**	0.868	0.870	0.501	0.736	0.750	0.848	0.745	0.778
TGADHead(Ours)	**0.884**	0.881	**0.746**	0.673	**0.682**	**0.877**	0.915	0.803	**0.634**	0.784	0.717	**0.797**	0.888	**0.875**	**0.873**	**0.533**	**0.765**	0.730	0.836	0.762	**0.780**

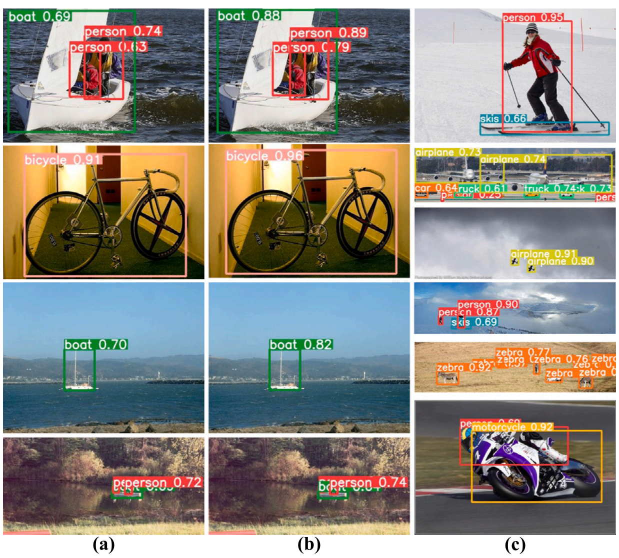

Fig. 11. Visualization of results. (a) and (b) respectively show the test results without COCO pre-training weight and the test results under pre-loading pre-training model. (c) shows the performance under COCO val2017 data set.

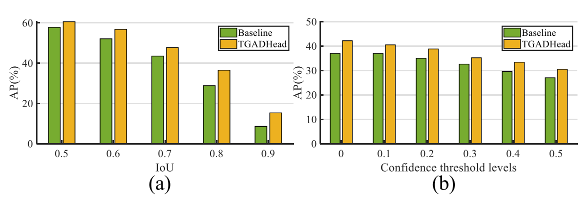

Fig. 12. Performance analysis. (a) shows the detection results of the two methods at IoU from 0.5 to 0.9. (b) shows the comparison of performance under the different confidence threshold levels.

Fig. 13. Visualization performance comparison between baseline model (top row) and TGADhead (bottom row).

Table 10
Comparison of operation efficiency under different detector head structures. Our method is light and prone to equip.

Method	BackBone	Parameters	GFLOPs
YOLOx-Head [55]	Modified CSP-s	11.4 M	27.8
PPYOLOE-Head [62]	Modified CSP-s	9.7 M	23.6
TOOD-Head [32]	Modified CSP-s	11.6 M	29.2
YOLOv6-Head [56]	Modified CSP-s	11.2 M	28.6
YOLOv7-Head [78]	Modified CSP-s	12.3 M	29.5
TGADHead(Ours)	Modified CSP-s	11.0 M	26.5

Table 11
AP across scale criteria from 0.5 to 0.9 with 0.1 interval.

Criteria	TGAD	$AP_{0.5}$	$AP_{0.6}$	$AP_{0.7}$	$AP_{0.8}$	$AP_{0.9}$
AP_s		38.6	34.1	26.6	16.7	3.8
AP_s	✓	**41.1**	**36.7**	**28.5**	**18.8**	**5.8**
AP_m		63.5	58.5	49.9	33.9	9.1
AP_m	✓	**68.1**	**63.7**	**55.6**	**41.3**	**15.9**
AP_l		70.0	66.1	56.9	43.5	15.2
AP_l	✓	**75.7**	**71.8**	**66.8**	**54.3**	**28.6**

4.5.2. Analyze the existing attention mechanism

To further demonstrate the performance proposed TGAD. We have introduced two plug-and-play modules [42,44] into the detection head. Fig. 14 shows the performance comparison of AP and $AP_{0.5}$ in different backbone networks using YOLO baseline. It can be found that our method has powerful performance when deployed in the detector compared with these attention models. The main reason is that we have purposely built task-specific attention allocators for the model prediction mechanism, which is different from some plug-and-play modules. Finally, the visualization results of the relevant object's attention map are shown in Fig. 15.

4.5.3. Assembled to other single-stage detectors

As the key component of the detector, the prediction network (Head) is responsible for generating object probability and location. In order to demonstrate the generalizability of TGADHead, we assembled the head to other mainstream single-stage detectors (RetinaNet, FCOS and ATSS) and compared the performance with their original prediction mechanisms. As shown in Table 12, the introduction of TGADHead in different detectors can effectively improve the overall performance, especially in the strict IoU threshold, the overall AP value increased by 3.5%, 3.4% and 3.9% respectively.

5. Conclusions

In this paper, we propose a novel Task-Guided Attention-Decoupled Head (TGADHead) to improve single-stage object detection accuracy.

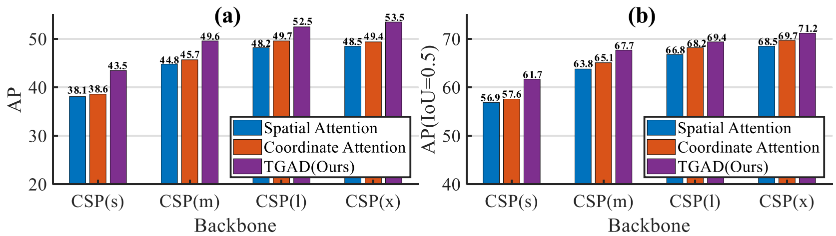

Fig. 14. Performance comparison in different attention. (a) and (b) show the performance comparison of AP and $AP_{0.5}$ with different attention mechanisms under diverse backbone networks.

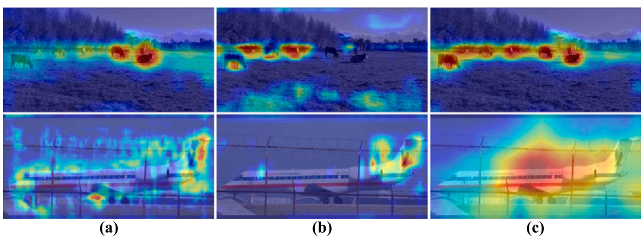

Fig. 15. Visual comparison of attention map. (a): Spatial Attention (SA), (b) Coordinate Attention (CA), (c): TGAD (Ours).

Table 12
Performance comparison of TGADHead assembled on different detectors (RetinaNet [50], FCOS [34], ATSS [54]).

Detector	Original	TGADHead	AP	$AP_{0.5}$	$AP_{0.75}$
RetinaNet	✓		40.8	61.1	44.1
RetinaNet		✓	**44.3**	**63.7**	**47.1**
FCOS	✓		41.5	60.7	45.0
FCOS		✓	**44.9**	**63.3**	**48.1**
ATSS	✓		47.7	66.5	51.9
ATSS		✓	**51.6**	**69.7**	**56.4**

Firstly, TDAD gathers favorable features for classification and localization by constructing two task specific attention blocks. This divide and conquer strategy avoids the entanglement between the two task features. Secondly, the TCN establishes the connection between the classification and the localization task, and finally completes the accurate object detection through adaptive weight assignment of various sub-tasks to cooperate with each other. A large number of experiments have verified the effectiveness of TGADHead and demonstrated dramatic performance improvement compared with the state-of-the-art methods. Our findings are generic in nature and can be used for general object detection applications.

CRediT authorship contribution statement

Fengyuan Zuo: Methodology. **Jinhai Liu:** Supervision, Methodology, Funding acquisition. **Zhaolin Chen:** Visualization. **Mingrui Fu:** Conceptualization. **Lei Wang:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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References

[1] F. PÍęrez-HernÍćndez, S. Tabik, A. Lamas, R. Olmos, H. Fujita, F. Herrera, Object detection binary classifiers methodology based on deep learning to identify small objects handled similarly: Application in video surveillance, Knowl.-Based Syst. 194 (2020) 105590.

[2] J. Nie, Z. Dong, Z. He, H. Wu, M. Gao, FAML-RT: Feature alignment-based multi-level similarity metric learning network for a two-stage robust tracker, Inform. Sci. 632 (2023) 529–542.

[3] W. Ma, T. Zhou, J. Qin, Q. Zhou, Z. Cai, Joint-attention feature fusion network and dual-adaptive NMS for object detection, Knowl.-Based Syst. 241 (2022) 108213.

[4] L. Yan, Y. Qin, J. Chen, Scale-balanced real-time object detection with varying input-image resolution, IEEE Trans. Circuits Syst. Video Technol. 33 (1) (2023) 242–256, http://dx.doi.org/10.1109/TCSVT.2022.3198329.

[5] Y. Xie, W. Wang, M. Shao, T. Li, Y. Yu, Multi-view change point detection in dynamic networks, Inform. Sci. 629 (2023) 344–357.

[6] S. Sun, N. Akhtar, H. Song, A. Mian, M. Shah, Deep affinity network for multiple object tracking, IEEE Trans. Pattern Anal. Mach. Intell. 43 (1) (2021) 104–119, http://dx.doi.org/10.1109/TPAMI.2019.2929520.

[7] H. Zhang, W. Xing, Y. Yang, Y. Li, D. Yuan, SiamST: Siamese network with spatio-temporal awareness for object tracking, Inform. Sci. 634 (2023) 122–139.

[8] F. Ma, M.Z. Shou, L. Zhu, H. Fan, Y. Xu, Y. Yang, Z. Yan, Unified transformer tracker for object tracking, in: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2022, pp. 8771–8780, http://dx.doi.org/10.1109/CVPR52688.2022.00858.

[9] F. Zhang, R. Wang, F. Zhou, Y. Luo, ERM: Energy-based refined-attention mechanism for video question answering, IEEE Trans. Circuits Syst. Video Technol. 33 (3) (2023) 1454–1467, http://dx.doi.org/10.1109/TCSVT.2022.3212463.

[10] F. Ma, X. Jin, H. Wang, J. Huang, L. Zhu, J. Feng, Y. Yang, Temporal perceiving video-language pre-training, 2023, ArXiv abs/2301.07463 URL https://api.semanticscholar.org/CorpusID:255998340.

[11] X. Shen, J. Liu, J. Sun, L. Jiang, H. Zhao, H. Zhang, SSCT-Net: A semisupervised circular teacher network for defect detection with limited labeled multiview MFL samples, IEEE Trans. Ind. Inform. (2023) 1–11, http://dx.doi.org/10.1109/TII.2022.3232764.

[12] Y. Ren, J. Liu, H. Zhang, W. Wang, TDDA-Net: A transitive distant domain adaptation network for industrial sample enhancement, Inform. Sci. 606 (2022) 927–944.

[13] H. Wang, Y. Sun, X. Bi, Structural redundancy reduction based efficient training for lightweight person re-identification, Inform. Sci. 637 (2023) 118962.

[14] Y. Cheng, Z. Wang, W. Zhan, H. Duan, Multi-scale human-object interaction detector, IEEE Trans. Circuits Syst. Video Technol. 33 (4) (2023) 1827–1838, http://dx.doi.org/10.1109/TCSVT.2022.3216663.

[15] Y. Xu, C. Zhou, X. Yu, Y. Yang, Cyclic self-training with proposal weight modulation for cross-supervised object detection, IEEE Trans. Image Process. 32 (2023) 1992–2002, http://dx.doi.org/10.1109/TIP.2023.3261752.

[16] Y. Shi, A. Li, Z. Deng, Q. Yan, Q. Lou, H. Chen, K.-S. Choi, S. Wang, Double-coupling learning for multi-task data stream classification, Inform. Sci. 613 (2022) 494–506.

[17] F. Ma, Y. Wu, X. Yu, Y. Yang, Learning with noisy labels via self-reweighting from class centroids, IEEE Trans. Neural Netw. Learn. Syst. 33 (11) (2022) 6275–6285, http://dx.doi.org/10.1109/TNNLS.2021.3073248.

[18] J. Han, X. Yao, G. Cheng, X. Feng, D. Xu, P-CNN: Part-based convolutional neural networks for fine-grained visual categorization, IEEE Trans. Pattern Anal. Mach. Intell. 44 (2) (2022) 579–590, http://dx.doi.org/10.1109/TPAMI.2019.2933510.

[19] Z. Huang, J. Wang, X. Fu, T. Yu, Y. Guo, R. Wang, DC-SPP-YOLO: Dense connection and spatial pyramid pooling based YOLO for object detection, Inform. Sci. 522 (2020) 241–258.

[20] Q. Wang, Y. Qian, Y. Hu, C. Wang, X. Ye, H. Wang, M2YOLOF: Based on effective receptive fields and multiple-in-single-out encoder for object detection, Expert Syst. Appl. 213 (2023) 118928.

[21] H. Qiu, H. Li, Q. Wu, J. Cui, Z. Song, L. Wang, M. Zhang, CrossDet++: Growing crossline representation for object detection, IEEE Trans. Circuits Syst. Video Technol. 33 (3) (2023) 1093–1108, http://dx.doi.org/10.1109/TCSVT.2022.3211734.

[22] Z. Wang, W. Zhu, W. Zhao, L. Xu, Balanced one-stage object detection by enhancing the effect of positive samples, IEEE Trans. Circuits Syst. Video Technol. (2023) 1, http://dx.doi.org/10.1109/TCSVT.2023.3237826.

[23] G. Cheng, X. Yuan, X. Yao, K. Yan, Q. Zeng, X. Xie, J. Han, Towards large-scale small object detection: Survey and benchmarks, IEEE Trans. Pattern Anal. Mach. Intell. 45 (11) (2023) 13467–13488, http://dx.doi.org/10.1109/TPAMI.2023.3290594.

[24] ultralytics, Yolov5, 2021, https://github.com/ultralytics/yolov5.

[25] R.R. Selvaraju, Cogswell, Grad-CAM: Visual explanations from deep networks via gradient-based localization, in: 2017 IEEE International Conference on Computer Vision, ICCV, 2017, pp. 618–626, http://dx.doi.org/10.1109/ICCV.2017.74.

[26] M. Hajizadeh, M. Sabokrou, A. Rahmani, MobileDenseNet: A new approach to object detection on mobile devices, Expert Syst. Appl. 215 (2023) 119348.

[27] X. Wei, S. Liu, Y. Xiang, Z. Duan, C. Zhao, Y. Lu, Incremental learning based multi-domain adaptation for object detection, Knowl.-Based Syst. 210 (2020) 106420.

[28] L. Yang, Y. Xu, S. Wang, C. Yuan, Z. Zhang, B. Li, W. Hu, PDNet: Toward better one-stage object detection with prediction decoupling, IEEE Trans. Image Process. 31 (2022) 5121–5133, http://dx.doi.org/10.1109/TIP.2022.3193223.

[29] Y. Lu, X. Chen, Z. Wu, J. Yu, Decoupled metric network for single-stage few-shot object detection, IEEE Trans. Cybern. 53 (1) (2023) 514–525, http://dx.doi.org/10.1109/TCYB.2022.3149825.

[30] L. Zhou, H. Chang, B. Ma, S. Shan, Interactive regression and classification for dense object detector, IEEE Trans. Image Process. 31 (2022) 3684–3696, http://dx.doi.org/10.1109/TIP.2022.3174391.

[31] H. Wang, Q. Wang, H. Zhang, Q. Hu, W. Zuo, CrabNet: Fully task-specific feature learning for one-stage object detection, IEEE Trans. Image Process. 31 (2022) 2962–2974, http://dx.doi.org/10.1109/TIP.2022.3162099.

[32] C. Feng, Y. Zhong, Y. Gao, M.R. Scott, W. Huang, TOOD: Task-aligned one-stage object detection, in: 2021 IEEE/CVF International Conference on Computer Vision, ICCV, 2021, pp. 3490–3499, http://dx.doi.org/10.1109/ICCV48922.2021.00349.

[33] F. Gao, Y. Cai, F. Deng, C. Yu, J. Chen, Feature alignment in anchor-free object detection, IEEE Trans. Circuits Syst. Video Technol. (2023) 1, http://dx.doi.org/10.1109/TCSVT.2023.3241993.

[34] Z. Tian, C. Shen, H. Chen, T. He, FCOS: A simple and strong anchor-free object detector, IEEE Trans. Pattern Anal. Mach. Intell. 44 (4) (2022) 1922–1933, http://dx.doi.org/10.1109/TPAMI.2020.3032166.

[35] Y. Sun, L. Su, Y. Luo, H. Meng, Z. Zhang, W. Zhang, S. Yuan, IRDCLNet: Instance segmentation of ship images based on interference reduction and dynamic contour learning in foggy scenes, IEEE Trans. Circuits Syst. Video Technol. 32 (9) (2022) 6029–6043, http://dx.doi.org/10.1109/TCSVT.2022.3155182.

[36] F. Ma, L. Zhu, Y. Yang, Weakly supervised moment localization with decoupled consistent concept prediction, Int. J. Comput. Vis. 130 (5) (2022) 1244–1258, http://dx.doi.org/10.1007/s11263-022-01600-0.

[37] J. Cao, Y. Pang, J. Han, X. Li, Hierarchical regression and classification for accurate object detection, IEEE Trans. Neural Netw. Learn. Syst. 34 (5) (2023) 2425–2439, http://dx.doi.org/10.1109/TNNLS.2021.3106641.

[38] Z. Chen, H. Ji, Y. Zhang, Z. Zhu, High-resolution feature pyramid network for small object detection on drone view, IEEE Trans. Circuits Syst. Video Technol. (2023) 1, http://dx.doi.org/10.1109/TCSVT.2023.3286896.

[39] G. Lin, S. Li, Y. Chen, X. Li, IDNet: Information decomposition network for fast panoptic segmentation, IEEE Trans. Image Process. (2023) 1, http://dx.doi.org/10.1109/TIP.2023.3234499.

[40] H. Wang, T. Jia, B. Ma, Q. Wang, W. Zuo, Fully cascade consistency learning for one-stage object detection, IEEE Trans. Circuits Syst. Video Technol. (2023) 1, http://dx.doi.org/10.1109/TCSVT.2023.3263557.

[41] M.-H. Guo, C. Lu, Z.-N. Liu, M.-M. Cheng, S. Hu, Visual attention network, 2022, ArXiv abs/2202.09741.

[42] Q. Hou, D. Zhou, J. Feng, Coordinate attention for efficient mobile network design, in: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2021, pp. 13708–13717, http://dx.doi.org/10.1109/CVPR46437.2021.01350.

[43] J. Hu, L. Shen, S. Albanie, et al., Squeeze-and-excitation networks, IEEE Trans. Pattern Anal. Mach. Intell. 42 (8) (2020) 2011–2023, http://dx.doi.org/10.1109/TPAMI.2019.2913372.

[44] X. Li, L. Huang, Z. Wei, A twofold convolutional regression tracking network with temporal and spatial mechanism, IEEE Trans. Circuits Syst. Video Technol. 32 (3) (2022) 1537–1551, http://dx.doi.org/10.1109/TCSVT.2021.3077640.

[45] K. Zhu, J. Wu, Residual attention: A simple but effective method for multi-label recognition, in: 2021 IEEE/CVF International Conference on Computer Vision, ICCV, 2021, pp. 184–193, http://dx.doi.org/10.1109/ICCV48922.2021.00025.

[46] J. Liu, X. Shen, An intelligent defect detection approach based on cascade attention network under complex magnetic flux leakage signals, IEEE Trans. Ind. Electron. (2022) 1–11, http://dx.doi.org/10.1109/TIE.2022.3201320.

[47] H. Li, S. Wang, R. Wan, A.C. Kot, GMFAD: Towards generalized visual recognition via multilayer feature alignment and disentanglement, IEEE Trans. Pattern Anal. Mach. Intell. 44 (3) (2022) 1289–1303, http://dx.doi.org/10.1109/TPAMI.2020.3020554.

[48] Y. Xu, Y. Sun, Z. Yang, J. Miao, Y. Yang, H2FA R-CNN: Holistic and hierarchical feature alignment for cross-domain weakly supervised object detection, in: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2022, pp. 14309–14319, http://dx.doi.org/10.1109/CVPR52688.2022.01393.

[49] Q. Chen, Y. Wang, T. Yang, X. Zhang, J. Cheng, J. Sun, You only look one-level feature, in: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2021, pp. 13034–13043, http://dx.doi.org/10.1109/CVPR46437.2021.01284.

[50] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. DollÍćr, Focal loss for dense object detection, IEEE Trans. Pattern Anal. Mach. Intell. 42 (2) (2020) 318–327, http://dx.doi.org/10.1109/TPAMI.2018.2858826.

[51] S.-Y. Wang, Z. Qu, C.-J. Li, A dense-aware cross-splitNet for object detection and recognition, IEEE Trans. Circuits Syst. Video Technol. 33 (5) (2023) 2290–2301, http://dx.doi.org/10.1109/TCSVT.2022.3221658.

[52] S. Zhang, L. Wen, Z. Lei, S.Z. Li, RefineDet++: Single-shot refinement neural network for object detection, IEEE Trans. Circuits Syst. Video Technol. 31 (2) (2021) 674–687, http://dx.doi.org/10.1109/TCSVT.2020.2986402.

[53] C. Shu, Y. Liu, J. Gao, et al., Channel-wise knowledge distillation for dense prediction, in: 2021 IEEE/CVF International Conference on Computer Vision, ICCV, 2021, pp. 5291–5300, http://dx.doi.org/10.1109/ICCV48922.2021.00526.

[54] S. Zhang, C. Chi, Y. Yao, Z. Lei, S.Z. Li, Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection, in: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2020, pp. 9756–9765, http://dx.doi.org/10.1109/CVPR42600.2020.00978.

[55] Z. Ge, S. Liu, et al., YOLOX: Exceeding YOLO series in 2021, 2021, ArXiv abs/2107.08430.

[56] C. Li, L. Li, H. Jiang, K. Weng, Y. Geng, L. Li, Z. Ke, Q. Li, M. Cheng, W. Nie, Y. Li, B. Zhang, Y. Liang, L. Zhou, X. Xu, X. Chu, X. Wei, X. Wei, YOLOv6: A single-stage object detection framework for industrial applications, 2022, ArXiv abs/2209.02976.

[57] H. Zhang, Y. Wang, F. Dayoub, N. SÍźnderhauf, VarifocalNet: An IoU-aware dense object detector, in: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2021, pp. 8510–8519, http://dx.doi.org/10.1109/CVPR46437.2021.00841.

[58] X. Chu, A. Zheng, Detection in crowded scenes: One proposal, multiple predictions, in: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2020, pp. 12211–12220, http://dx.doi.org/10.1109/CVPR42600.2020.01223.

[59] G. Cheng, P. Lai, D. Gao, J. Han, Class attention network for image recognition, Sci. China Inf. Sci. 66 (3) (2023) http://dx.doi.org/10.1007/s11432-021-3493-7.

[60] X. Xie, C. Lang, S. Miao, G. Cheng, K. Li, J. Han, Mutual-assistance learning for object detection, IEEE Trans. Pattern Anal. Mach. Intell. 45 (12) (2023) 15171–15184, http://dx.doi.org/10.1109/TPAMI.2023.3319634.

[61] Y. Liu, Z. Yu, D. Zong, L. Zhu, Attention to task-aligned object detection for EndÍCEdgeÍCCloud video surveillance, IEEE Internet Things J. 11 (8) (2024) 13781–13792, http://dx.doi.org/10.1109/JIOT.2023.3340151.

[62] S. Xu, X. Wang, W. Lv, Q. Chang, et al., PP-YOLOE: An evolved version of YOLO, 2022, ArXiv abs/2203.16250.

[63] P.-O. Bagnaninchi, M. Dikeakos, T. Veres, M. Tabrizian, Complex permittivity measurement as a new noninvasive tool for monitoring in vitro tissue engineering and cell signature through the detection of cell proliferation, differentiation, and pretissue formation, IEEE Trans. NanoBiosci. 3 (4) (2004) 243–250, http://dx.doi.org/10.1109/TNB.2004.837901.

[64] J. Luo, Z. Yang, S. Li, Y. Wu, FPCB surface defect detection: A decoupled two-stage object detection framework, IEEE Trans. Instrum. Meas. 70 (2021) 1–11, http://dx.doi.org/10.1109/TIM.2021.3092510.

[65] K.W. Lau, L.-M. Po, Y.A.U. Rehman, Large separable kernel attention: Rethinking the large kernel attention design in CNN, Expert Syst. Appl. (ISSN: 0957-4174) 236 (2024) 121352, http://dx.doi.org/10.1016/j.eswa.2023.121352.

[66] M.-H. Guo, C.-Z. Lu, Z.-N. Liu, M.-M. Cheng, S.-M. Hu, Visual attention network, 2022, arXiv:2202.09741.

[67] M. Hu, Y. Li, L. Fang, S. Wang, A2-FPN: Attention aggregation based feature pyramid network for instance segmentation, in: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2021, pp. 15338–15347, http://dx.doi.org/10.1109/CVPR46437.2021.01509.

[68] X. Yang, C. Deng, T. Liu, D. Tao, Heterogeneous graph attention network for unsupervised multiple-target domain adaptation, IEEE Trans. Pattern Anal. Mach. Intell. 44 (4) (2022) 1992–2003, http://dx.doi.org/10.1109/TPAMI.2020.3026079.

[69] Z. Zheng, P. Wang, D. Ren, W. Liu, R. Ye, Q. Hu, W. Zuo, Enhancing geometric factors in model learning and inference for object detection and instance segmentation, IEEE Trans. Cybern. 52 (8) (2022) 8574–8586, http://dx.doi.org/10.1109/TCYB.2021.3095305.

[70] X. Li, C. Lv, W. Wang, G. Li, L. Yang, J. Yang, Generalized focal loss: Towards efficient representation learning for dense object detection, IEEE Trans. Pattern Anal. Mach. Intell. 45 (3) (2023) 3139–3153, http://dx.doi.org/10.1109/TPAMI.2022.3180392.

[71] J. Wang, W. Zhang, Y. Cao, K. Chen, J. Pang, T. Gong, Side-aware boundary localization for more precise object detection, in: Computer Vision – ECCV 2020, 2020, pp. 403–419.

[72] Z. Cai, N. Vasconcelos, Cascade R-CNN: Delving into high quality object detection, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2018, pp. 6154–6162, http://dx.doi.org/10.1109/CVPR.2018.00644.

[73] H. Qiu, Y. Ma, Z. Li, S. Liu, J. Sun, BorderDet: Border feature for dense object detection, in: ECCV, 2020, pp. 549–564.

[74] Y. Wu, Y. Chen, L. Yuan, Z. Liu, L. Wang, H. Li, Y. Fu, Rethinking classification and localization for object detection, in: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2020, pp. 10183–10192, http://dx.doi.org/10.1109/CVPR42600.2020.01020.

[75] C. Zhu, Y. He, M. Savvides, Feature selective anchor-free module for single-shot object detection, in: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2019, pp. 840–849, http://dx.doi.org/10.1109/CVPR.2019.00093.

[76] G. Song, Y. Liu, X. Wang, Revisiting the sibling head in object detector, in: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2020, pp. 11560–11569, http://dx.doi.org/10.1109/CVPR42600.2020.01158.

[77] Y. Li, E. Zhu, H. Chen, J. Tan, L. Shen, Dense crosstalk feature aggregation for classification and localization in object detection, IEEE Trans. Circuits Syst. Video Technol. 33 (6) (2023) 2683–2695, http://dx.doi.org/10.1109/TCSVT.2022.3218880.

[78] C.-Y. Wang, A. Bochkovskiy, H.-Y.M. Liao, YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors, 2022, arXiv preprint arXiv:2207.02696.